

Coupled differential equations

Basic skills

Q1. $\frac{dy}{dx} = 3x + 2y$ $\frac{d^2y}{dx^2} = 3 + 2 \frac{dy}{dx}$

Q2. $\frac{d^2y}{dt^2} - \frac{dy}{dt} - 6y = 0$

aux: $m^2 - m - 6 = 0$ $(m-3)(m+2) = 0$

so $m = -2, 3$ so $y = Ae^{-2t} + Be^{3t}$

Q3. $\frac{dy}{dt} = x - 2y$, $\frac{dx}{dt} = x + y$

a) $\frac{d^2y}{dt^2} = \frac{dx}{dt} - 2 \frac{dy}{dt} = x + y - 2 \frac{dy}{dt}$

now $x = \frac{dy}{dt} + 2y$ so

$$\frac{d^2y}{dt^2} = 6y - \frac{dy}{dt} \Rightarrow \frac{d^2y}{dt^2} + \frac{dy}{dt} - 6y = 0$$

$$\text{So } m^2 + m - 6 = 0 \quad (m+3)(m-2) = 0$$

$$m = -3, 2 \quad y(t) = Ae^{-3t} + Be^{2t}$$

$$b) \text{ As } x = \frac{dy}{dt} + 2y \quad \text{need } \frac{dy}{dt}, y$$

$$\frac{dy}{dt} = -3Ae^{-3t} + 2Be^{2t}$$

$$\text{So } x = -3Ae^{-3t} + 2Be^{2t} + 2(Ae^{-3t} + Be^{2t})$$

$$x = -Ae^{-3t} + 4Be^{2t}$$

$$c) \quad y(0) = 0, \quad x(0) = 1$$

$$\Rightarrow A + B = 0, \quad -A + 4B = 1$$

$$\Rightarrow 5B = 1 \quad B = \frac{1}{5} \quad A = -\frac{1}{5}$$

$$\text{So } x = \frac{1}{5} (4e^{2t} - e^{-3t})$$

$$y = \frac{1}{5} (e^{2t} - e^{-3t})$$

d)



e) both x and y increase as time increases

As $t \rightarrow \infty$ $x \rightarrow \frac{4}{5}e^{2t}$ $y \rightarrow \frac{1}{5}e^{2t}$

So $y \rightarrow \infty$ along $y = \frac{1}{4}x$

Competency skills

Q4. a) $\frac{dy}{dt} = 2x - 3y$, $\frac{dx}{dt} = -2x + y$

$$\begin{aligned} \frac{d^2y}{dt^2} &= 2 \frac{dx}{dt} - 3 \frac{dy}{dt} = 2(-2x + y) - 3 \frac{dy}{dt} \\ &= -4x + 2y - 3 \frac{dy}{dt} = -2 \left(\frac{dy}{dt} + 3y \right) + 2y - 3 \frac{dy}{dt} \end{aligned}$$

$x = \frac{1}{2} \left(\frac{dy}{dt} + 3y \right)$

$$\text{so } \frac{d^2y}{dt^2} = -4y - 5\frac{dy}{dt}$$

$$\frac{d^2y}{dt^2} + 5\frac{dy}{dt} + 4y = 0$$

$$\text{aux } m^2 + 5m + 4 = 0 \Rightarrow (m+4)(m+1) = 0$$

$$\text{so } m = -1, -4 \text{ so } y = Ae^{-4t} + Be^{-t}$$

$$\text{Then } x = \frac{1}{2} \left(\frac{dy}{dt} + 3y \right) \text{ as } \frac{dy}{dt} = -4Ae^{-4t} - Be^{-t}$$

$$\begin{aligned} x &= \frac{1}{2} (-Ae^{-4t} + 2Be^{-t}) \\ &= -\frac{A}{2}e^{-4t} + Be^{-t} \end{aligned}$$

$$\text{Now } t=0, x=1, y=1$$

$$\text{so } -\frac{A}{2} + B = 1, A + B = 1$$

$$\text{so } A = 2B - 2 \Rightarrow 3B - 2 = 1 \Rightarrow B = 1$$

$$\text{and } A = 0 \text{ so } x = e^{-t}, y = e^{-t}$$

$$\text{so } y = x$$


Linear increase in x results in linear in y

b) $\frac{dy}{dt} = x - 2y - 5, \quad \frac{dx}{dt} = y - 3x + 4$

so $\frac{d^2y}{dt^2} = \frac{dx}{dt} + 2 \frac{dy}{dt} = y - 3x + 4 - 2 \frac{dy}{dt}$

as $x = \frac{dy}{dt} + 2y + 5$

$\Rightarrow \frac{d^2y}{dt^2} = y - 3 \left(\frac{dy}{dt} + 2y + 5 \right) + 4 - 2 \frac{dy}{dt}$
 $= -5y - 11 - 5 \frac{dy}{dt}$

so $y'' + 5y' + 5y = -11$

so aux $m^2 + 5m + 5 = 0$

$$m = \frac{-5 \pm \sqrt{25 - 4 \times 25}}{2}$$

$m = -\frac{5}{2} \pm \frac{5\sqrt{3}}{2} i$ so $y = e^{-\frac{5}{2}t} \left(A \cos\left(\frac{5\sqrt{3}}{2}t\right) + B \sin\left(\frac{5\sqrt{3}}{2}t\right) \right) - \frac{11}{5}$

PI: need $5y = -11$ so $-\frac{11}{5}$ as well

so $\frac{dy}{dt} = -\frac{5}{2} e^{-\frac{5}{2}t} \left(A \cos\left(\frac{5\sqrt{3}}{2}t\right) + B \sin\left(\frac{5\sqrt{3}}{2}t\right) \right)$
 $+ \frac{5\sqrt{3}}{2} e^{-\frac{5}{2}t} \left(-A \sin\left(\frac{5\sqrt{3}}{2}t\right) + B \cos\left(\frac{5\sqrt{3}}{2}t\right) \right)$

Then

$$x = \frac{dy}{dt} + 2y + 5$$

$$= -\frac{1}{2}e^{-\frac{s}{2}t} \left(A \cos\left(\frac{s\sqrt{3}}{2}t\right) + B \sin\left(\frac{s\sqrt{3}}{2}t\right) \right) + \frac{s\sqrt{3}}{2}e^{-\frac{s}{2}t} \left(-A \sin\left(\frac{s\sqrt{3}}{2}t\right) + B \cos\left(\frac{s\sqrt{3}}{2}t\right) \right) + \frac{3}{5}$$

Applying $x=0, y=2$ at $t=0$

$$\Rightarrow -\frac{1}{2}A + \frac{s\sqrt{3}}{2}B + \frac{3}{5} = 0$$

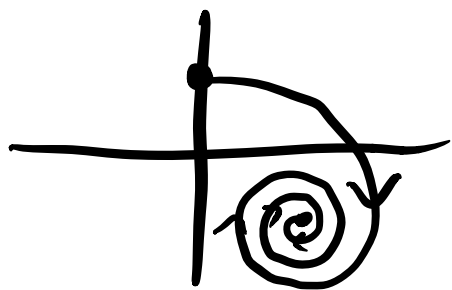
$$A = 2$$

$$\text{so } B = \frac{\frac{1}{2} \times 2 - \frac{3}{5}}{\frac{s\sqrt{3}}{2}} = \frac{\frac{2}{5}}{s\sqrt{3}} = \frac{2\sqrt{3}}{75}$$

$$\text{so } x = e^{-\frac{s}{2}t} \left(-\frac{4}{5} \cos\left(\frac{s\sqrt{3}}{2}t\right) - \frac{376\sqrt{3}}{75} \sin\left(\frac{s\sqrt{3}}{2}t\right) \right) + \frac{3}{5}$$

$$y = e^{-\frac{s}{2}t} \left(2 \cos\left(\frac{s\sqrt{3}}{2}t\right) + \frac{2\sqrt{3}}{75} \sin\left(\frac{s\sqrt{3}}{2}t\right) \right) - \frac{11}{5}$$

as $t \rightarrow \infty$ $x \rightarrow \frac{3}{5}, y \rightarrow -\frac{11}{5}$
so spirals around $(\frac{3}{5}, -\frac{11}{5})$



Q5.

$$\frac{dA}{dt} = 2B - 9A + 14 \quad \frac{dB}{dt} = 7A - 3B$$

a) +14 means chemical A is being constantly added at this rate 14

b) For no reaction $\frac{dA}{dt} = 0$, $\frac{dB}{dt} = 0$

$$\text{so } 2B - 9A + 14 = 0, \quad 7A - 3B = 0$$

$$\Rightarrow 9A - 2B = 14$$

$$7A - 3B = 0 \Rightarrow B = \frac{7}{3}A$$

$$\text{so } 9A - \frac{14}{3}A = 14 \quad \frac{13}{3}A = 14$$

$$A = \frac{42}{13} \quad B = \frac{7}{3} \times \frac{42}{13} = \frac{98}{13}$$

$$\begin{aligned} \text{c) } \frac{d^2A}{dt^2} &= 2 \frac{dB}{dt} - 9 \frac{dA}{dt} \\ &= 2(7A - 3B) - 9 \frac{dA}{dt} = 14A - 6B - 9 \frac{dA}{dt} \end{aligned}$$

as

$$B = \frac{1}{2} \left(\frac{dA}{dt} + 9A - 14 \right)$$

$$A'' = 14A - 3A' - 27A + 42 - 9A'$$

$$\Rightarrow A'' + 12A' + 13A = 42$$

$$\text{aux } m^2 + 12m + 13 = 0$$

$$m = \frac{-12 \pm \sqrt{12^2 - 4 \times 13}}{2} = -6 \pm \sqrt{23}$$

$$\text{also need } 13A = 42 \text{ so } A = \frac{42}{13}$$

$$\text{so } A = a e^{(-6+\sqrt{23})t} + b e^{(-6-\sqrt{23})t} + \frac{42}{13}$$

$$\frac{dA}{dt} = (-6+\sqrt{23})a e^{(-6+\sqrt{23})t} + (-6-\sqrt{23})b e^{(-6-\sqrt{23})t}$$

$$B = \frac{1}{2} \left(\begin{aligned} &(-6+\sqrt{23})a e^{(-6+\sqrt{23})t} \\ &+ (-6-\sqrt{23})b e^{(-6-\sqrt{23})t} \\ &+ 9a e^{(-6+\sqrt{23})t} + 9b e^{(-6-\sqrt{23})t} \end{aligned} + 9 \times \frac{42}{13} - 14 \right)$$

$$= \frac{3+\sqrt{23}}{2} a e^{(-6+\sqrt{23})t} + \frac{3-\sqrt{23}}{2} b e^{(-6-\sqrt{23})t} + \frac{98}{13}$$

$$\text{as } t \rightarrow \infty, A \rightarrow \frac{42}{13}, B \rightarrow \frac{98}{13} \quad \leftarrow \text{like with } A' = 0 = B'$$

Q6.

a) $20R' = 2R - w$, $100w' = R$

Then $100w'' = R' = \frac{1}{20}(2R - w)$

$$2000w'' = 2R - w$$

$$= 2 \times 100w' - w$$

$$2000w'' - 200w' + w = 0$$

aux: $m^2 - \frac{1}{10}m + \frac{1}{2000} = 0$

$$m = \frac{5 \pm 2\sqrt{5}}{100}$$

so $w = Ae^{\frac{5+2\sqrt{5}}{100}t} + Be^{\frac{5-2\sqrt{5}}{100}t}$

$$R = 100w' = (5+2\sqrt{5})Ae^{\frac{5+2\sqrt{5}}{100}t} + (5-2\sqrt{5})Be^{\frac{5-2\sqrt{5}}{100}t}$$

b) Let $t=0$, $R=2000$, $w=30$

then $30 = A + B$, $2000 = (5+2\sqrt{5})A + (5-2\sqrt{5})B$

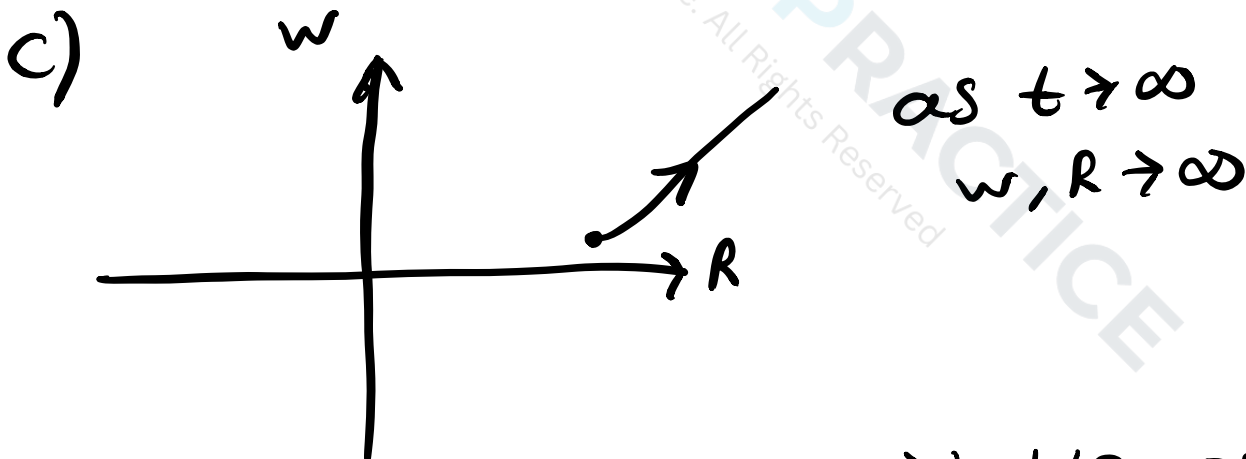
so $A = 30 - B \Rightarrow 2000 = 30(5+2\sqrt{5}) - 4\sqrt{5}B$

$$\text{so } B = \frac{1850 - 60\sqrt{5}}{-4\sqrt{5}} = 15 - \frac{185\sqrt{5}}{2}$$

$$A = 30 - \left(15 - \frac{185\sqrt{5}}{2}\right) = 15 + \frac{185\sqrt{5}}{2}$$

$$\text{so } w = \left(15 + \frac{185\sqrt{5}}{2}\right) e^{\frac{5+2\sqrt{5}}{100}t} + \left(15 - \frac{185\sqrt{5}}{2}\right) e^{\frac{5-2\sqrt{5}}{100}t}$$

$$R = (5+2\sqrt{5}) \left(15 + \frac{185\sqrt{5}}{2}\right) e^{\frac{5+2\sqrt{5}}{100}t} + (5-2\sqrt{5}) \left(15 - \frac{185\sqrt{5}}{2}\right) e^{\frac{5-2\sqrt{5}}{100}t}$$



d) model is not suitable as have both exponentially grow forever